GinCol Lab · Aula

Student worksheet

Mathematics I · 1st year of Bachillerato · The real numbers · Class 4 of 6 · With Ximena

Intervals and absolute value: distances on the number line

Class video: https://youtu.be/MF3P3M3Lt9g

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. In Spain the decimal separator is a comma: the board writes 19,9 for 19.9, and that is why it separates the endpoints of an interval with a semicolon, [19,9; 20,1]. On this sheet, with the decimal point, the interval is written [19.9, 20.1].

Before watching the video

1. A part has to measure 20 mm, "plus or minus a tenth of a millimetre". Tick ✓ the ones that pass:

20.07	19.97	20.13	19.85	20.1	19.9

Were you unsure about any of them? Why?

2. Work out: 7 − 3 = _____ and 3 − 7 = _____. How far is it from 3 to 7 on the number line? And from 7 to 3? Which calculation gives you the distance in both cases?

3. Your bet: what is the highest age a person who is a minor can be, measured in years with all its decimals?

While you watch the video

1. Between two different numbers, is there always room for another? How do you find it?

2. **Your prediction.** Does a part that measures exactly 20.1 pass? Why?

3. Write the interval of the parts that pass. Why does the board write it with a semicolon?

4. Write as intervals the age of a minor and the age of an adult. Why does the 18 have a square bracket in one and a round bracket in the other?

5. **Your bet.** What is the largest number in [0, 18)? Why doesn't 17.999... count?

6. Why does infinity always have a round bracket?

7. What does the sign of a subtraction tell you, and what does its absolute value tell you?

8. Why does the rule for the absolute value write $-x$, if the absolute value is never negative?

9. Which number is the same distance from 1 as from 5? What does it have to do with the first question?

10. What happens to the window $|x - a| \leq r$ if the radius grows? And if it is 0? And with $|x - a| < 0$?

11. Some people solve |x| < 3 by dropping the bars: "x < 3". Which number takes that apart, and what is the right solution?

The four kinds. Complete the table.

Inequality	Drawing on the line	Interval	It is called
19.9 ≤ x ≤ 20.1			
2 < x < 5			
0 ≤ x < 18			
x < 3			

The window, with gaps

Write the interval [–4, 10] with an absolute value, filling in the gaps without looking at the video.

1 · The centre. It is the midpoint of the endpoints: (_____ + _____) / 2 = _____.

2 · The radius. It is half the length of the interval: (_____ – _____) / 2 = _____.

3 · The absolute value. |x – _____| _____ _____. Why with ≤ and not with <?

Now, in your own words:

1. Check your answer with –4, with 10 and with 11: do they satisfy it or not?

2. How would you write the open interval (–4, 10)? What is that interval called?

Practice

1. Write $|x - 2| \leq 0.5$ as an interval. Are $\sqrt{2}$ and $\sqrt{5}$ inside it? Hint: compare their squares.

2. Write the interval $[-1, 3]$ with an absolute value.

3. Which numbers satisfy $|x - 1| > 2$? Write it with intervals.

4. **One more, to revise class 1.** Is $[0, 1)$ the same as $[0, 0.999\ldots]$? Why?

5. **One more, to revise class 3.** Is π in the interval $(3.14, 22/7)$? What is its centre?

6. **A challenge.** A part is made at 50 mm with a tolerance of ± 0.2 mm. Write the ones that pass with an absolute value, and the ones that do not with intervals. What is the largest distance between two parts that pass?

Keep thinking

No measurement comes out with radius zero: every measurement has its margin, and it is an interval. So what does it mean that something measures exactly two metres? Does an exact measurement exist?

GinCol Lab worksheet, made with AI (Claude, by Anthropic) and reviewed by a person · CC BY 4.0 · Source and errata